

STAT 5210 Research Proposal: Analyzing Firm-Level Innovation and Spillover Effects Using Econometric Methods

Libin Chen

1. Introduction

Technological innovation is a key driver of economic growth and firm competitiveness. This study aims to extend the existing research on the relationship between R&D expenditure, patent applications, and technological spillovers. We will investigate questions such as: the causal impact of technological spillovers on firm innovation while controlling for potential endogeneity in R&D investment and spillovers; the dynamic effects of past innovation performance on current R&D investment.

2. Dataset

The study will use the Cincera (1997) dataset, which includes 181 international manufacturing firms that invested significantly in R&D during the late 1980s. The dataset covers European patent applications, R&D expenditures, technological spillovers, and firm characteristics from 1983 to 1991.

(Link: <https://journaldata.zbw.eu/dataset/patents-rampd-and-technological-spillovers-at-the-firm-level-some-evidence-from-econometric-count-m>)

3. Analysis methods

We will employ econometric techniques such as System 2SLS, GMM, and potentially First Differencing methods for the estimation. System 2SLS will be used to address potential endogeneity in R&D and spillovers. System GMM will be used to model the dynamic effects of past innovation performance on current R&D investment. And we will employ the First Differencing (FD) method to eliminate firm-specific fixed effects of the firm itself, to better estimate the causal relationship between the number of patents and R&D expenditures. We can further verify the relationship between patents and R&D expenditures by combining System 2SLS or GMM with FD, so the influence of additional factors such as technology spillover and industry cycle is considered in the model.